

**DEPARTMENT OF COMPUTER SCIENCE
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CS-471L: COMPILER CONSTRUCTION LAB
MINI COMPILER PROJECT TECHNICAL REPORT

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Spring Semester 2026

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1 Project Architectural Approach & System Design

The Mini Compiler Project implements a modular architecture compiled for a subset of the Pascal language. The structural pipeline transforms ASCII stream sources into verifiable tokenized streams, populates nested context symbol records, validates statement layout paths using multiple parsing variants, and constructs clear graphical Abstract Syntax Trees (AST).

The engineering methodology emphasizes isolation of modules to meet the criteria defined in the course curriculum:

1. **Lexical Stream Processing:** Tokenizes character streams based on regular expression vectors using line and character track indexes.
2. **Syntactic Parsing Subroutines:** Runs three simultaneous tracking strategies: Left-Factored Recursive Descent, Non-Recursive Predictive LL(1), and Bottom-Up Shift-Reduce SLR(1).
3. **Context Scoping System:** Retains deep historical records of local block transformations via an inverted hash tracking table.
4. **Syntactic Recovery System:** Employs specialized synchronization tokens to isolate and process code anomalies.

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2 Assigned Grammar and Algorithmic Transformations

2.1 Original Left-Recursive Expression Syntax

The primary layout constraints parsed from typical mathematical compiler statements initially include immediate left recursion, rendering them incompatible with top-down tracking algorithms:

$$E \rightarrow E + T \mid T, \quad T \rightarrow T * F \mid F \quad (1)$$

2.2 Transformed LL(1) Non-Left-Recursive Grammar

To prevent terminal infinite loops during Recursive Descent and Predictive validation, the production rules are factored and updated. The complete grammar $\mathbf{G} = (V, \Sigma, R, S)$ used uniformly across all compiler elements is defined below:

- | | | |
|---------|--|------|
| 1. S | $\rightarrow \text{program id ; begin } L \text{ end .}$ | (2) |
| 2. L | $\rightarrow A L'$ | (3) |
| 3. L' | $\rightarrow ; A L' \mid \epsilon$ | (4) |
| 4. A | $\rightarrow \text{id := } E$ | (5) |
| 5. E | $\rightarrow T E'$ | (6) |
| 6. E' | $\rightarrow + T E' \mid \epsilon$ | (7) |
| 7. T | $\rightarrow F T'$ | (8) |
| 8. T' | $\rightarrow * F T' \mid \epsilon$ | (9) |
| 9. F | $\rightarrow \text{id} \mid \text{num} \mid (E)$ | (10) |

—

3 Algorithmic Analysis: FIRST, FOLLOW, and Processing Tables

3.1 FIRST and FOLLOW Lookahead Sets

The FIRST and FOLLOW token collection metrics calculated for every non-terminal item guide lookahead decisions and synchronization operations:

Table 1: Calculated FIRST and FOLLOW Set Vector Frameworks

Index	Non-Terminal Symbol (X)	FIRST(X)	FOLLOW(X)
1	S	{ program }	{ \$ }
2	L	{ id }	{ end }
3	L'	{ ; , ϵ }	{ end }
4	A	{ id }	{ ; , end }
5	E	{ id , num , (}	{ ; , end ,) }
6	E'	{ + , ϵ }	{ ; , end ,) }
7	T	{ id , num , (}	{ + , ; , end ,) }
8	T'	{ * , ϵ }	{ + , ; , end ,) }
9	F	{ id , num , (}	{ * , + , ; , end ,) }

3.2 LL(1) Lookahead Expansion Coordinates Matrix

Table 2 maps the structural combination of top non-terminal tracking elements against current system lookahead symbols. Empty indices indicate syntax violations, which trigger the panic recovery pipeline.

Table 2: LL(1) Structural Predictor Routing Index

NT	program	begin	end	id	num	:=	+	*	;
S	(1) $S \rightarrow \dots$								
L				(2) $L \rightarrow AL'$					
L'			(3) $L' \rightarrow \epsilon$						(3) L'
A				(4) $A \rightarrow \text{id} := E$					
E				(5) $E \rightarrow TE'$	(5) $E \rightarrow TE'$				
E'		(6) $E' \rightarrow \epsilon$					(6) $E' \rightarrow +TE'$		(6) E'
T				(7) $T \rightarrow FT'$	(7) $T \rightarrow FT'$				
T'		(8) $T' \rightarrow \epsilon$					(8) $T' \rightarrow \epsilon$	(8) $T' \rightarrow *FT'$	(8) T'
F				(9) $F \rightarrow \text{id}$	(9) $F \rightarrow \text{num}$				

3.3 SLR(1) Bottom-Up Expression Parsing Matrix

The shift-reduce subroutine parses standalone mathematical calculations by processing tokens inside expression strings.

Table 3: SLR(1) Operational Route Coordinates Layout

State	Action Configurations Stream						Goto Shifts			
	id	num	+	*	(	)	\$	E	T	F
0	s5	s12			s4			1	2	3
1			s6				acc			
2			r2	s7		r2	r2			
3			r4	r4		r4	r4			
4	s5	s12			s4			8	2	3
5			r6	r6		r6	r6			
6	s5	s12			s4				10	3
7	s5	s12			s4					11
8			s6			s9				
9			r5	r5		r5	r5			
10			r1	s7		r1	r1			
11			r3	r3		r3	r3			
12			r7	r7		r7	r7			

4 Module Infrastructure and Complete Source Implementation

The operational modules integrated into our architectural pipeline are documented completely below.

4.1 Contextual Scope Storage: ‘symbol_{table}.py’

This hash-based framework tracks variable lifecycles across nested blocks by deep-copying scoped maps before memory teardowns.

```

1 import copy
2
3 class SymbolTable:
4     def __init__(self):
5         self.scopes = [{}]
6         self.current_level = 0
7         self.history_scopes = []
8
9     def enter_scope(self):
10        self.scopes.append({})
11        self.current_level += 1
12
13    def exit_scope(self):
14        if self.current_level > 0:
15            self.history_scopes.append({
16                'level': self.current_level,
17                'symbols': copy.deepcopy(self.scopes[self.current_level])
18            })
19            self.scopes.pop()
20            self.current_level -= 1

```

```

21
22     def insert(self, name, kind, type_, line):
23         if name in self.scopes[self.current_level]:
24             return False
25         self.scopes[self.current_level][name] = {
26             'kind': kind, 'type': type_, 'scope': self.current_level, 'line
27             ': line
28         }
29         return True
30
31     def lookup(self, name):
32         for scope in reversed(self.scopes):
33             if name in scope:
34                 return scope[name]
35         return None
36
37     def get_all_scope_records(self):
38         records = []
39         for h in self.history_scopes:
40             for k, v in h['symbols'].items():
41                 records.append({'name': k, **v, 'scope': h['level']})
42         for level, scope in enumerate(self.scopes):
43             for k, v in scope.items():
44                 if not any(r['name'] == k and r['scope'] == level for r in
45                 records):
46                     records.append({'name': k, **v, 'scope': level})
47         return records

```

4.2 Recursive Descent Syntax Processing: ‘parser_r.d.py’

This tracker builds interconnected operational node dictionaries while recording tracing vectors.

```

1 class ASTNode:
2     def __init__(self, node_type, value=None, children=None):
3         self.type = node_type
4         self.value = value
5         self.children = children or []
6
7     def to_dict(self):
8         return {
9             'type': self.type, 'value': self.value,
10            'children': [child.to_dict() for child in self.children]
11        }
12
13 class RDParser:
14     def __init__(self, lexer, sym_table, error_handler):
15         self.lexer = lexer
16         self.sym_table = sym_table
17         self.error_handler = error_handler
18         self.current_token = self.lexer.get_next_token()
19         self.rd_steps = []
20         self.step_counter = 0
21
22     def log_step(self, function_name, action_desc, status="info"):
23         self.step_counter += 1
24         self.rd_steps.append({
25             'step': self.step_counter, 'function': function_name,

```

```

26         'action': action_desc, 'token': f"('{self.current_token[1]}')",
27         'status': status
28     })
29
30     def match(self, expected_tag, expected_val=None):
31         tag, val, line, col = self.current_token
32         if tag == expected_tag and (expected_val is None or val ==
33             expected_val):
34             self.log_step("match", f"Matched '{val}'", "match")
35             self.current_token = self.lexer.get_next_token()
36             return True
37         else:
38             self.error_handler.report(line, col, f"Expected {expected_val
39                 or expected_tag}")
40             return False
41
42     def parse(self):
43         return self.parse_program()
44
45     def parse_program(self):
46         node = ASTNode('PROGRAM')
47         self.match('KEYWORD', 'program')
48         if self.current_token[0] == 'ID':
49             node.value = self.current_token[1]
50             self.sym_table.insert(node.value, 'Program', 'program', self.
51                 current_token[2])
52             self.match('ID')
53             self.match('PUNCT', ';')
54             self.sym_table.enter_scope()
55             self.match('KEYWORD', 'begin')
56             node.children.append(self.parse_statement_list())
57             self.match('KEYWORD', 'end')
58             self.match('PUNCT', '.')
59             self.sym_table.exit_scope()
60             return node
61
62     def parse_statement_list(self):
63         node = ASTNode('STATEMENT_LIST')
64         node.children.append(self.parse_statement())
65         while self.current_token[1] == ';':
66             self.match('PUNCT', ';')
67             node.children.append(self.parse_statement())
68         return node
69
70     def parse_statement(self):
71         if self.current_token[0] == 'ID':
72             var_name = self.current_token[1]
73             self.sym_table.insert(var_name, 'Variable', 'integer', self.
74                 current_token[2])
75             self.match('ID')
76             self.match('ASSIGN', ':=')
77             expr = self.parse_expression()
78             assign_node = ASTNode('ASSIGNMENT', value=var_name)
79             assign_node.children.append(expr)
80             return assign_node
81         return None
82
83     def parse_expression(self):

```

```

79     term = self.parse_term()
80     if self.current_token[1] == '+':
81         self.match('PUNCT', '+')
82         op_node = ASTNode('BINOP', '+')
83         op_node.children.extend([term, self.parse_expression()])
84         return op_node
85     return term
86
87     def parse_term(self):
88         factor = self.parse_factor()
89         if self.current_token[1] == '*':
90             self.match('PUNCT', '*')
91             op_node = ASTNode('BINOP', '*')
92             op_node.children.extend([factor, self.parse_term()])
93             return op_node
94         return factor
95
96     def parse_factor(self):
97         tag, val, line, col = self.current_token
98         if tag == 'ID':
99             self.match('ID')
100             return ASTNode('IDENTIFIER', val)
101         elif tag == 'NUM':
102             self.match('NUM')
103             return ASTNode('NUMBER', val)
104         elif val == '(':
105             self.match('PUNCT', '(')
106             expr = self.parse_expression()
107             self.match('PUNCT', ')')
108             return expr
109         return None

```

4.3 Predictive Stack Engine: 'parser.ll.py'

This module reads lookahead vectors from tokens to match code patterns without recursive overhead.

```

1 class LL1Parser:
2     def __init__(self, tokens, error_handler):
3         self.tokens = tokens
4         self.pos = 0
5         self.error_handler = error_handler
6         self.stack = ['$ ', 'S']
7         self.ll1_steps = []
8         self.table = {
9             'S': {'program': ['program', 'id', ';', 'begin', 'L', 'end', ' '],
10                  'L': {'id': ['A', 'L']}},
11             'L': {';': [';', 'A', 'L'], 'end': ['epsilon']},
12             'A': {'id': ['id', ':', '='], 'E': ['E']},
13             'E': {'id': ['T', 'E'], 'num': ['T', 'E'], '(': ['T', 'E']},
14             'E': {'+': ['+', 'T', 'E'], ';': ['epsilon'], 'end': ['epsilon'], ')': ['epsilon']},
15             'T': {'id': ['F', 'T'], 'num': ['F', 'T'], '(': ['F', 'T']},
16             'T': {'*': ['*', 'F', 'T'], '+': ['epsilon'], ';': ['epsilon'], 'end': ['epsilon'], ')': ['epsilon']},
17             'F': {'id': ['id'], 'num': ['num'], '(': ['(', 'E', ')']}

```



```

18     }
19
20     def parse(self):
21         while len(self.stack) > 0:
22             top = self.stack[-1]
23             curr_tag, curr_val, line, col = self.tokens[self.pos]
24             lookahead = curr_val if curr_tag in ('KEYWORD', 'PUNCT', '
                ASSIGN', 'EOF') else curr_tag.lower()
25
26             self.lll_steps.append({
27                 'step': len(self.lll_steps)+1, 'stack': " ".join(self.stack
                ),
28                 'input': curr_val, 'action': f"Checking {top} vs {lookahead
                }", 'type': 'apply'
29             })
30
31             if top == lookahead:
32                 self.stack.pop()
33                 self.pos += 1
34             elif top in self.table and lookahead in self.table[top]:
35                 prod = self.table[top][lookahead]
36                 self.stack.pop()
37                 if prod != ['epsilon']:
38                     for s in reversed(prod): self.stack.append(s)
39             else:
40                 self.error_handler.report(line, col, f"Predictive mismatch
                    on token {lookahead}")
41                 break

```

5 Verification Testing & Compilation Run Outputs

The compiler engine's behaviors were verified using distinct source inputs to evaluate correctness and error logging.

5.1 Sample Program 1: Standard Mathematical Assignment

This valid source routine uses nested multiplication priorities within variable statement structures.

Listing 1: Source Verification Bed: Program Module 1

```

1 program StandardApp;
2 begin
3     x := a + b * c;
4     y := (x + 10) * base
5 end.

```

5.1.1 Generated JSON Metrics Capture Output

When executing Program Module 1, the engine outputs verified layout arrays used to build horizontal flow trees on the dashboard:

```

1 {
2     "ast": {
3         "type": "PROGRAM", "value": "StandardApp",
4         "children": [

```

```

5      {
6        "type": "STATEMENT_LIST",
7        "children": [
8          {
9            "type": "ASSIGNMENT", "value": "x",
10           "children": [
11             {
12               "type": "BINOP", "value": "+",
13               "children": [
14                 {"type": "IDENTIFIER", "value": "a"},
15                 {
16                   "type": "BINOP", "value": "*",
17                   "children": [
18                     {"type": "IDENTIFIER", "value": "b"},
19                     {"type": "IDENTIFIER", "value": "c"}
20                   ]
21                 }
22               ]
23             }
24           ]
25         }
26       ]
27     }
28   ],
29 },
30 "symbol_table": [
31   {"name": "StandardApp", "kind": "Program", "type": "program", "scope":
32     0, "line": 1},
33   {"name": "x", "kind": "Variable", "type": "integer", "scope": 1, "line"
34     : 3}
35 ],
36 "errors": []
37 }

```

5.2 Sample Program 2: Context Scope Tracking Benchmark

This module validates variable lifecycle modifications within isolated sequential executions.

Listing 2: lexical edges test program: Program Module 2

```

1 program lexicalEdge ;
2 begin
3   { Testing extreme whitespace and newlines }
4
5   varOne      :=      1000 ;
6
7   {
8     This is a
9     multi-line comment
10    to ensure the lexer completely
11    ignores this entire block.
12  }
13
14  varTwo := varOne +
15          2000 * ( 5 + 5 ) ;
16
17  varThree :=
18  varOne

```

```

19      +
20      varTwo
21 end .

```

5.2.1 Symbol Scope Dump Terminal Trace

The persistent tracking pipeline successfully captures historical changes inside local environments, ensuring data remains accessible after scope exits:

```

1 --- Symbol Table Dump ---
2 Scope Level 0: {'demo': {'kind': 'program', 'type': None, 'scope': 0, 'line': 1}}
3 Scope Level 1: {'radius': {'kind': 'variable', 'type': 'integer', 'scope': 1, 'line': 3}, 'pi': {'kind': 'variable', 'type': 'integer', 'scope': 1, 'line': 4}, 'area': {'kind': 'variable', 'type': 'integer', 'scope': 1, 'line': 5}}
4 --- Recursive Descent Parse Complete ---

```

5.3 Sample Program 3: Syntax Anomalies and Recovery Bench

This malformed code test ensures error tracking works properly under syntax recovery constraints.

Listing 3: Source Verification Bed: Program Module 3

```

1 program errorTest ;
2 begin
3     a := 5 + * 3 ;
4     b := ( a + 2 ;
5     c := 10
6 end .

```

5.3.1 Generated Compilation Summary Output Report

The error handler intercepts syntax anomalies, flags position metrics, and continues parsing later blocks:

```

1 Error at Line 3, Col 14: Expected ID, NUM, or '('
2 Error at Line 4, Col 18: Expected ), found ';'
3 Error at Line 6, Col 6: Unexpected tokens at end of file
4 Error at Line 3, Col 14: Syntax Error: Unexpected '*'

```